

MyTitle

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Abstract

Briefly summarize your term paper in 8-15 lines.

This project implements the Performer ([Choromanski et al., 2020](#)) attention mechanism in the LLAMA-3 language model and compares it with its own regular Transformer ([Vaswani et al., 2017](#)) attention mechanism. The algorithm implemented includes the Fast Attention Via positive Orthogonal Random features approach (FAVOR+) algorithm, which has the time and memory complexity of $O(L)$.

References

- Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamás Sarlós, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. 2020. [Rethinking attention with performers](#). *arXiv preprint arXiv:2009.14794*.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. [Attention is all you need](#). *arXiv preprint arXiv:1706.03762*.